

Somya Agarwal

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology - Computer Science and Engineering

October 2022 – May 2026

CGPA: 8.63

WORK EXPERIENCE

AI/ML Intern, NSpark Software LLC

January 2025 – March 2025

- Engineered a scalable Python data-ingestion backend with modular connectors to onboard data from multiple enterprise sources and built secure OAuth 2.0 integrations (Google Drive/Sheets), enabling governed access to external systems for automated pipelines.
- Designed and productionized REST APIs and service-layer architecture for ingestion workflows, improving maintainability, extensibility, and turnaround time for onboarding new integrations.

Machine Learning Intern, Suvidha Foundation

May 2025 – July 2025

- Developed an AI pipeline combining **credibility analysis** and **BERT-based summarization** models, flagged over 500 unreliable news articles per day while reducing average summarization processing time by **15%**, enabling faster content assessment.
- Achieved a **+35%** improvement in trust score and reduced fake news content in summaries by over 50% through rigorous evaluation.

Software Developer Intern, Deeporion Technology

October 2024 – December 2024

- Developed a real-time sentiment analysis engine (**AngularJS, Java, and MySQL**), processing 25k+ customer reviews daily; reduced feedback loop by **40%**.
- Improved **MongoDB** and **PostgreSQL** queries across 10+ apps, reducing retrieval time by 35 seconds using optimized indexing and joins.

SKILLS AND EXPERTISE

Languages	: Python, JAVA, JavaScript, TypeScript
AI/ML	: LLMs, RAG, BERT, Vector Embeddings, Generative AI
Databases	: MySQL, PostgreSQL, PostGIS, SQLite, MongoDB, Drizzle
Data & Visualization	: Pandas, NumPy, Matplotlib, Seaborn, Statistics
Cloud & Devops	: Docker, CI/CD Pipelines, GIT/GitHub, Rest APIs
Frameworks	: React.js, Node.js, Express.js, FastAPI, Spring Boot
CS Fundamentals	: Data Structures & Algorithms, OOP, Operating Systems

ACADEMIC PROJECTS

CrisisWatch – AI-Powered Crisis Mapping Platform [\[Live\]](#)

- Designed a full-stack crisis mapping platform with **React, FastAPI, and PostgreSQL/POSTGIS**, enabling visualization of real-time disaster incidents with **1.5-3 seconds latency**.
- Implemented a robust **PostgreSQL + Alembic migration system**, ensuring 100% database schema version control and smooth **CI/CD** pipelines.

AI-Powered Enterprise Knowledge Base and RAG Chatbot

- Built an end-to-end document **ingestion** and **metadata enrichment pipeline**, combining user-provided column descriptions with LLM-based semantic analysis.
- Designed and implemented an **AI-Powered Knowledge Base system** to convert structured CSV datasets into semantically searchable intelligence using LLMs and Retrieval-Augmented Generation (RAG).

Skill Navigator - AI-Powered Learning Platform [\[Live\]](#)

- Designed and managed **PostgreSQL** data models with Drizzle ORM, including migration-driven schema evolution for assessments, roadmap steps, and ML tracking.
- Developed progress analytics features to track completion status, time spent, and estimated learning timelines.

CERTIFICATIONS

- Full Stack Development from Apna college | Coursera Frontend Certification - John Hopkin's University
- GEN AI (IBM) | Java Developer Professional (OCP)

ACHIEVEMENTS

Smart India Hackathon (SIH)

- Led a team of 4 in developing a prototype ranked in top 5 out of 200 submissions; contributed to backend and ML modules.

HackOnBlocks

- Directed a standout prototype ranked as one of the top 5 projects out of 150+ teams, demonstrating strong design and execution.